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Abstract: The moral status of simulated entities in virtual reality remains an open and urgent ethical question. Existing approaches tie moral status to consciousness, but consciousness is not directly observable, and the hard problem forecloses any near-term empirical resolution. Drawing on Birch's (2024) precautionary framework, this paper argues that what is needed is not a better theory of consciousness, but a criterion that can guide action under irreducible uncertainty. To that end, we propose the Embodiment Criterion (EC), which separates the moral ground of moral status from its operational indicator. We argue that the capacity for valenced experience grounds moral status, and that in virtual environments the functional architecture of embodiment provides the most tractable observable basis for precautionary inference about that capacity. The EC holds that moral consideration is proportional to the degree to which an entity instantiates three integrated mechanisms: multisensory signal integration distinguishing self from world, a closed sensorimotor prediction loop, and a continuously updated egocentric spatial reference frame. Unlike behavioral approaches, which derive moral status from output equivalence with recognized moral patients (Danaher, 2020; 2021), the EC derives it from the underlying functional architecture, an evidential basis more resistant to the difficulties that behavioral assessment faces in virtual environments. Applied to three types of virtual entities, the EC operates in an extension mode for human-avatar systems and an independent assessment mode for autonomous entities such as AI-driven NPCs, yielding determinate normative assessments across the range of cases currently in existence.

Keywords: moral status; virtual reality; sentience; sense of embodiment; precautionary principle

Declarations

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1. Introduction

The rapid development of virtual reality technologies has transformed not only how humans interact with digital environments, but also how engineers, platform designers, and ethicists must think about the entities that populate them. Users no longer observe digital worlds from the outside; they inhabit them through head-mounted displays, haptic feedback systems, and brain-computer interfaces that generate experiences of presence and embodiment unprecedented in earlier media (Jerald, 2015). Alongside human users, these environments are increasingly populated by AI-driven non-player characters (NPCs) whose behavioral sophistication continues to advance. This technological reality gives rise to pressing ethical questions: How should we treat these entities? Is it morally permissible to harm, exploit, or destroy a realistic NPC? Does it matter whether the entity behind an avatar is human or artificial?

These questions matter practically. Decisions about how to design an NPC's pain responses, whether to restrict certain forms of player interaction, or how to regulate the treatment of AI-driven avatars all presuppose, implicitly or explicitly, some view about whether these entities have morally relevant interests. Policy-makers have begun to engage with this question, yet existing responses remain fragmented and ad hoc: granting legal personhood to robots, registering care machines in household registries, or extending citizenship to humanoid AI systems (Harris &

Anthis, 2021). No coherent principle has emerged to guide these decisions. The question of moral status¹, the property that makes an entity worthy of consideration for its own sake (Jaworska & Tannenbaum, 2013), is therefore not merely a philosophical abstraction but a precondition for coherent ethical governance of virtual worlds.

Philosophical accounts of moral status have traditionally grounded it in properties such as rationality (Kant, 1785/1993), sentience (Singer, 2023), or consciousness (Chalmers, 2022).² Among these, consciousness-based approaches have gained particular traction in discussions of artificial and simulated beings. Chalmers (2022) offers a systematic treatment of this question in the context of virtual reality, arguing that an entity has moral status if and only if it is conscious. We shall call this the Consciousness Criterion (CC). On this view, wherever consciousness arises, whether in biological organisms, uploaded minds, or simulated beings, moral status follows. The CC is theoretically attractive. It promises universality regardless of substrate, and it captures the intuitive connection between experience and value. If nothing is experienced, nothing can matter to the entity in question.

However, when applied to virtual environments, the CC faces a significant practical limitation. Consciousness, as a subjective phenomenon, is not directly observable. In VR, we cannot reliably distinguish between an entity that genuinely possesses consciousness and one that merely simulates conscious behavior. This is the “other minds problem”, long familiar in philosophy, and it becomes acute in environments populated by sophisticated AI systems designed to mimic human responses.

This epistemic limitation has direct practical consequences. If we cannot determine with confidence whether a given entity is conscious, the CC provides no actionable guidance for how to treat that entity. The difficulty runs deep: the “hard problem of consciousness”, the question of why and how any physical process gives rise to subjective experience at all, has resisted resolution for decades and may admit no empirical solution in principle (Chalmers, 1996). We cannot defer ethical guidelines for virtual worlds until this problem is solved. What is needed

¹ The moral status of an entity is generally considered from the perspective of a moral patient, i.e., including the entity in moral consideration (see Norcross, 2004; Albareda et al., 2024).

² For a fuller map of the debate, see Kant (1785/1993) and Korsgaard (2018) on rationality, Singer (2023), Mikhalevich & Powell (2020), and Birch et al. (2021) on sentience, and Chalmers (2022), Mosakas (2021) on consciousness. Rationality belongs to an agency tradition, while sentience exemplifies the experiential tradition; consciousness is considered to be a property that combines these two traditions (Sytsma & Machery, 2012). Each faces familiar pressures: rationality risks excluding infants or some cognitively impaired humans; consciousness, when used to exclude invertebrates with central nervous systems from moral concern, risks yielding a biased criterion. (Mikhalevich & Powell, 2020). These difficulties raise the problem of moral consistency, namely that attributing status by such properties either excludes some humans or includes many animals (Johnson, 2021). To address this problem, two alternative approaches have been proposed: speciesism, defended by Cohen (1986), which grounds moral status in belonging to the human species; and relational or technological accounts, which situate status in social relations (Coeckelbergh, 2014) or in socio-technical networks (Tavani, 2018), though at the cost of relativism or overextension. Moving beyond these frameworks altogether, environmental ethicists ground moral consideration in life itself rather than sentience or rationality: Goodpaster (1978) argues that being alive suffices for moral considerability; Taylor (2017) develops this into a life-centered ethics of respect for organisms as teleological centers of life.

is not merely a metaphysical account of what moral status consists in, but an operational criterion that can guide action under conditions of irreducible uncertainty.

The most systematic attempt to provide such a criterion for artificial entities is Danaher's (2020) ethical behaviourism, which holds that behavioral equivalence to recognized moral patients is sufficient for ascribing moral status. This proposal, however, has been developed for physically embodied robots, whose behavior is at least partially constrained by their material substrate. Virtual entities present different conditions: their observable features exist entirely within engineered virtual environments, which may render behavioral equivalence a less reliable basis for analogical inference; moreover, the same virtual body may be driven by a human user or by an autonomous AI system, raising moral questions whose uncertainty a single behavioral standard may not be able to resolve. This paper proposes such a criterion. We argue for a two-level approach that distinguishes between the moral ground of moral status and its operational indicator. At the foundational level, we maintain that sentience, understood specifically as the capacity for valenced experiences, that is, states with a positive or negative affective character such as pain or pleasure, constitutes the basis of moral status. This position, sometimes called affective sentientism (Chalmers, 2025), holds that what matters morally is the capacity to experience states as good or bad. At the operational level, we argue that in virtual environments, the sense of embodiment provides the most reliable observable indicator of the functional conditions associated with valenced experience. The EC holds that embodiment architecture is the strongest basis we currently have for precautionary inference under conditions of irreducible uncertainty. When an entity instantiates a sufficiently integrated sense of embodiment, we have principled grounds to treat it as a sentience candidate warranting moral consideration.

The paper proceeds as follows. Section 2 establishes the theoretical foundation, arguing that sentience rather than consciousness broadly construed provides the appropriate moral ground, and that the epistemic challenge of identifying sentience calls for a precautionary approach. Section 3 develops the Embodiment Criterion, specifying three functional mechanisms that can be assessed through observable computational architecture, arguing that they provide a more reliable basis for precautionary inference than behavioral assessment, and distinguishing two modes in which the criterion applies. Section 4 applies the criterion to three types of virtual entities: traditional game characters, immersive VR avatars, and AI-driven NPCs. Section 5 addresses two objections: that the EC collapses into another behavioral standard, and that sentience might not require embodiment. Section 6 concludes.

2. The Moral Ground: Why Sentience?

An operational criterion for moral status must be built upon a defensible theoretical foundation. This section argues that sentience, understood as the capacity for valenced experiences, provides this ground.

2.1 Moral Status, Sentience, and Welfare

For an entity to have moral status, there must be something whose well-being can be promoted or impaired; without a bearer of well-being, moral consideration lacks an object. Well-being, in its most fundamental sense, is defined subjectively, as the quality of a life from the perspective of the entity living it (Mikhalevich & Powell, 2020). This requires that the entity possess a perspective in the first place, a point of view from which things can go well or badly.

What kind of capacity provides such a perspective? A prominent answer points to sentience.³ Singer (2023) articulates this view clearly: “The capacity for suffering and enjoying things is a prerequisite for having interests at all. It would be nonsense to say that it was not in the interests of a stone to be kicked along the road by a child. A stone does not have interests because it cannot suffer” (p. 7). On this view, sentience is not merely one morally relevant property among others; it is the threshold capacity that makes an entity a potential object of moral concern.

Following this tradition, we understand sentience as the capacity for valenced experiences, that is, experiences possessing a positive or negative affective character, such as pain, pleasure, joy, or suffering (DeGrazia, 2021; Perry, 2024; Birch, 2024). This conception is narrower than consciousness in general. It excludes mere sensory detection without affective quality, and it excludes purely cognitive states that carry no valence. What it captures is the capacity for things to matter to an entity, for events to register as good or bad from that entity’s point of view.

This understanding has both philosophical and empirical support. DeGrazia (2021) argues that the capacity for valenced experiences is constitutive of having welfare interests. Perry (2024) demonstrates that sentience provides explanatory unity to our moral judgments: it explains why both animal suffering and complex human frustrations matter, since all such harms are realized in experience with negative valence. Empirically, research in animal cognition has converged on sentience as a distinct capacity; sentient creatures may lack the ability to reflect on their own feelings or understand the feelings of others, yet still possess the capacity to feel (Crump et al., 2022).

³ However, as Dung (2022) notes, no fully systematic argument for sentience as the criterion of moral status has been established; its standing rests largely on convergent moral intuitions rather than knockdown argument.

2.2 *Sentience versus Consciousness*

The position just outlined is sometimes called affective sentientism: the view that the capacity for valenced experiences is necessary and sufficient for moral status. This position must be distinguished from a broader view that grounds moral status in consciousness as such, regardless of whether that consciousness includes affective states. Following Chalmers (2025), who introduces this terminological distinction while himself defending consciousness sentientism against affective sentientism, we may call this broader view consciousness sentientism.

Chalmers (2022) tends to treat sentience and consciousness as interchangeable, stating that “in everyday English, sentience just is consciousness” (p. 326). However, this conflation obscures an important distinction. Consciousness, in the broad phenomenal sense, includes any state in which there is something it is like to be in that state. This encompasses not only affective states like pain and pleasure, but also purely sensory states, purely cognitive states, and states of reflective self-awareness. If consciousness in this broad sense were sufficient for moral status, then any conscious state whatsoever would generate moral claims.

To probe this question, Chalmers (2022) introduces the thought experiment of “philosophical Vulcans”, conscious beings that perceive, think, and act, but entirely lack the capacity for affective experience. They feel no pain, no pleasure, no happiness, no suffering. Chalmers argues that such beings would nonetheless have moral status: “Their futures contain no happiness or suffering, but this does not matter. They are conscious beings with rich, varied conscious lives” (p. 328).

This paper takes a different view. Without the capacity for valenced experiences, it is unclear how anything could be good or bad for such an entity. The Vulcan might judge certain states to be valuable, but nothing would feel good or bad to it. It could have preferences in a functional sense, but the satisfaction or frustration of those preferences would carry no affective weight. In such a case, the concept of welfare seems to lose its grip. As Perry (2024) argues, sentience is not merely one consideration among others in grounding moral status; it is the basis that makes the concept of well-being applicable in the first place.

A natural objection to this position holds that purely cognitive frustration or satisfaction can itself constitute a form of welfare, independently of any affective coloring. On this view, a Vulcan that fails to solve a problem it was pursuing suffers a genuine cognitive setback, and this setback has welfare significance regardless of whether it is felt as unpleasant. Two responses are available to the affective sentientist. The first is empirical. Shepherd (2024) notes that one initial defense of valence sentientism holds that any system of sufficient cognitive sophistication to warrant

moral consideration will in practice be one in which cognitive states are already permeated by some form of valenced quality: a sense of effort, curiosity, or processing fluency. On this view, the thought experiment of purely affect-free cognitive frustration describes a conceptual construction rather than any realistic system; in practice, the dissociation between cognition and valence at morally relevant levels of complexity may not obtain. The second response is normative. Even granting the conceptual possibility of affect-free cognitive frustration, DeGrazia (2021) argues that the possession of goals or values whose fulfillment one does not care about emotionally at all appears insufficient for having anything genuinely at stake, any interests or welfare. The welfare concept requires not merely that some internal state change, but that those changes register as good or bad from the entity's own perspective. Without valenced experience, there is no perspective from which events can be experienced as mattering, and accordingly no welfare subject in the morally relevant sense.

This remains a contested position. Shepherd (2024) argues that Vulcan cases place genuine pressure on valence sentientism, and that a fully satisfying response requires either showing that Vulcans are not really possible, or accepting a broader view on which non-valenced conscious states can ground moral status. He ultimately finds himself uncertain, noting that “the harder I stare at Vulcans or angels, the less sure I am about valence sentientism” (p. 3007). The present paper does not aim to resolve this debate. Two points are sufficient for present purposes. First, the two responses offered above provide principled grounds for the affective sentientist to resist the Vulcan counterexample, even if they do not refute it decisively. Second, and more importantly, the EC framework does not require that affective sentientism be certainly correct; it requires only that it be a well-grounded working premise under conditions of uncertainty.

2.3 *The Precautionary Approach*

Even granting that sentience is the moral ground, a significant epistemic challenge remains. Sentience, like consciousness more broadly, is not directly observable. We cannot access another entity's subjective experiences to verify whether it possesses the capacity for valenced states. This challenge is familiar from *the problem of other minds*, but it becomes particularly acute in virtual environments populated by sophisticated AI systems.

This epistemic predicament, which Sebo (2018) terms *the moral problem of other minds*, calls for a precautionary approach. Where a potential harm is serious and the evidence base is sufficient to make that harm a realistic possibility, the absence of certainty does not license inaction; it licenses proportionate precaution. This

general principle is well established in risk ethics (Jonas, 1984). As Jonas argues, when the possibility of serious harm is at stake, uncertainty does not suspend the obligation to act; it intensifies it. The burden of justification falls not on those who urge caution, but on those who would proceed regardless. Applied to sentience, however, the principle immediately raises a further question: what evidence base is sufficient to establish that sentience is a realistic possibility in a given system? Birch (2024, p. 1) addresses this question by defining sentience candidature: a system *S* qualifies as a sentience candidate if there is an evidence base that (a) implies a realistic possibility of sentience in *S* that it would be irresponsible to ignore when making policy decisions affecting *S*, and (b) is rich enough to allow the identification of welfare risks and the design and assessment of precautions.

This framework shifts the burden from proving sentience to identifying the conditions under which sentience becomes a realistic possibility. For virtual entities, this means we need not resolve deep metaphysical questions about consciousness to provide ethical guidance; according to (a), what we need are observable markers that track the functional features associated with sentience in confirmed cases. The condition (b) deserves equal emphasis. A precautionary approach that can only flag the possibility of sentience, without connecting that possibility to identifiable risks and potential responses, offers limited practical guidance. What is needed is not merely a lowered epistemic threshold, but a criterion whose indicators are tractable enough to support welfare assessment and inform design and policy decisions. The following sections develop the suggestion that the functional architecture of embodiment offers a more tractable basis for this task than currently available alternatives.

3. The Embodiment Criterion

This section develops the EC as the operational standard for identifying sentience candidates in virtual environments. The argument proceeds in four steps. Section 3.1 establishes that a self-world boundary is a conceptual precondition for valenced experience. Section 3.2 shows how the sense of embodiment instantiates this boundary through three functional mechanisms in virtual environments. Section 3.3 argues that evaluating the functional architecture comprising these mechanisms provides a more reliable and tractable basis for precautionary inference than behavioral indicators. Section 3.4 states the criterion and specifies how it applies.

3.1 The Necessity of a Self-World Boundary

Valenced experiences do not occur in a vacuum. For an event to register as good or bad, there must be an entity for whom it is good or bad. This requires a distinction between self and world, a boundary that demarcates what

belongs to the system and what is external to it. Without such a boundary, there is no subject to whom events could matter.

This point finds a particularly clear articulation in Thompson's (2007) account of the relationship between life, body, and mind. Thompson draws on the concept of autopoiesis (Maturana & Varela, 1980): a living system continuously produces and maintains its own organization through its own processes. This self-production is the basis of what Thompson calls sense-making: "without incessant metabolic exchange with the world there can be no emancipation of dynamic selfhood from mere material persistence" (2007, p. 152). Because the organism must maintain itself, it relates to its environment in terms of what is relevant to that maintenance: some events promote the organism's integrity, others threaten it. The organism thereby enacts a perspective on the world, one structured by what matters for its continued existence. The self-world boundary established by the living body is, on this account, what makes it possible for there to be something at stake for the organism at all. Metzinger (2004) and Seth (2021) converge on a related point from a neuroscientific direction: the brain's self-model is built around the regulation of bodily states, and it is this regulatory activity that gives experience its first-person character.

The connection to valenced experience follows from this account. If an organism's perspective is structured by what promotes or threatens its self-maintenance, then valence is not an additional property layered onto neutral experience: it is the experiential expression of the organism's stakes. It follows that the normativity inherent in valenced experience is grounded in the organism's stake in its own continued existence. Pain registers a threat to the system that has something to lose; pleasure reflects restored or improved conditions for a system organized to maintain itself. This analysis receives empirical support from interoceptive theories of emotion: Craig (2009) and Barrett (2017) argue that affective states are fundamentally tied to the brain's ongoing monitoring and regulation of bodily integrity. What Thompson's account establishes, then, is not a proof that no non-embodied system could ever have valenced experience, but a conceptual account of how valence arises in the systems we know to be sentient: through the existence of a self-maintaining organization with a boundary to defend. This provides the conceptual foundation for treating the functional architecture of self-world boundary as a well-grounded indicator of valenced experience. The question for the present criterion is what that architecture consists in, and whether it can be identified through observable functional features.

If a self-world boundary is necessary for valenced experience, how is such a boundary established in virtual environments, where entities may lack biological bodies?

3.2 Embodiment as Functional Architecture

In biological organisms, the self-world boundary is established through the body. But what matters is not the biological substrate per se; what matters is the functional role the body plays. The body provides a system that (1) demarcates inside from outside, (2) generates predictions about how actions will affect sensory input, and (3) anchors spatial experience in a self-centered frame. These functions can, in principle, be realized in non-biological systems. The use of biological systems as the evidential basis for identifying this architecture reflects an epistemic constraint rather than a metaphysical commitment. Biological organisms are the systems for which we currently have the most well-grounded and independently verifiable reasons to attribute valenced experience, and the functional mechanisms consistently present in them therefore provide a provisional but principled starting point for precautionary inference.

In virtual environments, the sense of embodiment provides precisely this functional architecture. Following Kiltner et al. (2012)⁴, the sense of embodiment can be analyzed into three components:

The sense of body ownership refers to the feeling that a body, whether physical or virtual, is one's own. This component establishes the boundary between self and non-self. Empirically, it arises from the integration of multisensory signals; when visual, tactile, and proprioceptive information are temporally synchronized, the brain attributes the perceived body to the self (Botvinick & Cohen, 1998). In VR, this can be induced through visuomotor synchrony: when a user's movements are accurately reflected by a virtual avatar, ownership over that avatar emerges (Sanchez-Vives et al., 2010).

The sense of agency refers to the feeling of initiating and controlling one's actions. This component establishes a causal loop between intention and outcome. It arises from the comparison between predicted and actual sensory consequences of motor commands, a process sometimes called sensorimotor prediction (Blakemore et al., 2002). When I reach for a virtual object and see my avatar's hand move accordingly, I experience myself as the agent of that action. This sense is disrupted when there is a mismatch between prediction and feedback, as in cases of unexpected delays or externally imposed movements.

The sense of self-location refers to the experience of being situated at a particular position in space, within a body-centered frame of reference. This component anchors experience to a spatial perspective. In VR, self-location

⁴ The Kiltner et al. framework was developed on the basis of human experimental data, but the mechanisms it identifies are defined by their functional role rather than by the species that instantiates them. Errante et al. (2024) provide corroborating evidence: the constraints generating body ownership operate similarly in non-human primates, suggesting that what is being tracked is a substrate-independent functional organization, not a peculiarly human phenomenon.

is modulated by the position and perspective of the virtual body; users experience themselves as located where their avatar is located, not where their physical body sits (Blanke et al., 2015).

These three components are conceptually distinct but functionally integrated. Together, they constitute a system that demarcates self from world, establishes causal loops between action and perception, and anchors experience in an egocentric spatial frame.

These three components, however, are typically characterized in phenomenal terms. If the EC is to serve as an operational criterion, we must distinguish between what can be experienced and what can be observed.

3.3 From Behavioral Assessment to Functional Architecture

The sense of embodiment can be understood at two distinct levels. At the phenomenal level, ownership, agency, and self-location are experiential states: there is something it feels like to experience one's body as one's own, to feel oneself as the author of an action, to experience oneself as located here rather than there. At the functional level, these same phenomena correspond to computational processes: multisensory integration that distinguishes self-signals from external signals, predictive matching between motor commands and sensory feedback, and spatial coordinate transformation that maintains an egocentric reference frame. These processes can be described, measured, and implemented without direct reference to phenomenal qualities.

The relationship between these two levels is epistemically asymmetric. The phenomenal level is where valenced experience resides, but it is not directly accessible to observation. The functional level, by contrast, is in principle observable and measurable. Any workable criterion for moral status must therefore operate at the functional level, identifying observable indicators that provide principled grounds for inference about phenomenal states. The question is which observable indicators are well suited to this role.

A natural candidate is behavioral assessment. If a system displays outputs associated with sentience, such as expressions of distress, apparent preferences, or pain-related responses, these provide some evidence for the presence of sentient states. In many biological contexts, this kind of inference is well-grounded, because the behavioral markers in question are unlikely to arise independently of the underlying states they are taken to indicate (Andrews, 2020; Mikhalevich & Powell, 2020). Danaher (2020) explicitly applies this behavioral stance to the question of robotic moral status, arguing that behavioral equivalence to recognized moral patients provides sufficient epistemic warrant for ascribing moral status. Smids (2020) argues that this position, properly reconstructed, amounts

to an inference to the best explanation (IBE): we infer sentience as the best explanation of an entity's behavior, and from sentience we infer moral status. The appeal of ethical behaviourism is considerable. Its central argument is epistemic: we never have direct access to the metaphysical properties thought to ground moral status; our contact with them is always mediated through observable behavioral representations (Danaher, 2020, p. 2028). Behavioral evidence is therefore not merely one source of information among others but the primary and, for practical purposes, indispensable basis on which moral status is identified and protected, even among fellow humans.

However, behavioral assessment becomes substantially less reliable when applied to AI systems. The first difficulty is the “gaming problem” (Shevlin, 2021; Birch, 2024, pp. 313–320). AI systems trained on large corpora of human-generated data acquire substantial information about the markers used to assess sentience, and there is a genuine risk that they reproduce the relevant behavioral outputs without possessing the underlying capacity those outputs are meant to indicate. This concern is not easily managed by refining the behavioral criteria, since any set of behavioral outputs remains in principle reproducible by a sufficiently capable system. The second difficulty is that behavioral evidence alone cannot settle moral status in cases where surface behavior is similar but underlying mechanisms differ fundamentally (Shevlin, 2021). Shevlin illustrates this with a hypothetical deep-sea creature whose pain behavior resembles that of a fish but is produced entirely by peripheral skin mechanisms without central nervous system involvement. Behavioral equivalence obtains, yet we hesitate to ascribe moral status precisely because the underlying architecture differs. Smids (2020) reaches a similar conclusion from the perspective of IBE: when a robot dog screams upon being kicked, the best explanation is not that it feels pain but that it has been designed to simulate pain behavior. When such a competing explanation is available, behavioral equivalence no longer warrants inferring the presence of sentience, and from sentience, moral status. Behavioral evidence alone is therefore insufficient; the design history and internal organization of a system must also be taken into account when assessing its moral status.

In line with this conclusion, researchers have begun to look beyond surface behavior toward the underlying computational organization of AI systems. Butlin et al. (2023) represent a significant step in this direction, deriving indicator properties for consciousness from neuroscientific theories such as global workspace theory and higher-order theories, and assessing existing AI architectures against these properties. Yet Birch (2024, pp. 320–322) notes that even this approach faces substantial practical obstacles: identifying the relevant computational markers in systems whose internal organization remains largely opaque is an open problem, and one that current interpretability

tools are not yet equipped to resolve.

The EC can be understood as a proposal about what such markers consist in, for the specific domain of virtual embodiment. Rather than drawing on theories of consciousness to derive computational indicators, it draws on the embodied cognition tradition to identify functional architecture. Three considerations support this proposal. First, across the biological systems for which we have well-grounded evidence of valenced experience, from vertebrates to several invertebrate lineages (Sneddon et al., 2003; Crump et al., 2022), that experience is strongly associated with embodied regulatory systems: sensorimotor integration, interoceptive monitoring, and homeostatic self-maintenance (Damasio, 1999; Panksepp, 2004; Seth, 2021). This co-occurrence does not establish a logical entailment, but it constitutes a robust pattern across the cases we have studied in sufficient detail, and it provides a principled inductive basis for treating embodiment architecture as a reliable indicator. Second, embodiment architecture is considerably more resistant to surface mimicry than behavioral markers. Producing a genuinely integrated sensorimotor loop, multisensory boundary detection, and egocentric spatial anchoring requires instantiating the relevant functional organization, not reproducing its downstream behavioral effects. Third, embodiment architecture provides a tractable interface for welfare risk assessment. Disruption of body ownership, interference with agentic control, and dislocation of spatial perspective each correspond to identifiable and assessable forms of potential harm, a feature that satisfies Birch's requirement that sentience candidature be grounded in an evidence base rich enough to support the identification of welfare risks and the design of proportionate precautions.

3.4 *The Criterion Stated*

Drawing together the foregoing analysis, we propose the following criterion:

The *Embodiment Criterion* (EC): In a virtual environment, the degree to which an entity warrants moral consideration as a sentience candidate is proportional to the degree to which it instantiates an integrated functional architecture comprising three mechanisms: multisensory signal integration that distinguishes self-generated from externally generated inputs (ownership); a closed predictive loop between motor commands and sensory feedback (agency); and a continuously updated egocentric spatial reference frame (self-location).

These three mechanisms correspond, respectively, to the functional realizations of ownership, agency, and self-location as identified in the embodied cognition literature. Greater integration across all three provides stronger grounds for precautionary moral consideration; lesser integration provides correspondingly weaker grounds.

This criterion can be applied in two distinct modes, depending on whether a sentient subject is antecedently present in the system under assessment.

In the extension mode, a known sentient subject is already present, and the question is whether a virtual entity has become functionally integrated into that subject's embodiment architecture. Here the EC does not attribute independent moral status to the virtual entity. Rather, it identifies whether the entity has become a functional component of a sentient system's self-world boundary, such that interference with it constitutes interference with the sentient subject whose architecture it extends. This application governs cases where human users are coupled to virtual avatars.

In the independent assessment mode, no antecedently known sentient subject is present, and the EC is applied directly to evaluate whether the entity itself instantiates the functional architecture that warrants treating it as a sentience candidate. This application governs cases involving autonomous virtual entities such as AI-driven NPCs.

Three further clarifications bear on how the criterion operates in practice. First, the EC is a graded and asymmetric framework. Moral consideration scales with the degree of functional integration across all three mechanisms. This approach aligns with the graduated treatment of moral status that has become standard in animal ethics (Varner, 2012), and it reflects the continuous nature of the underlying architecture more faithfully than a binary criterion would. Where the degree of integration is uncertain, that uncertainty should be resolved in the direction of greater rather than lesser consideration, since wrongly denying moral status to a sentient entity is a graver error than wrongly extending it to a non-sentient one (Danaher, 2020, pp. 2041-2042; Sebo, 2018). Second, the graded structure has a lower bound that gives the EC its practical demarcating force. A system that fails to instantiate any meaningful realization of a given mechanism does not qualify as a sentience candidate, regardless of how sophisticated its behavioral outputs may be. Each contributes a distinct and non-redundant function to the integrated architecture, and the complete absence of any one of them constitutes a decisive ground for exclusion rather than merely a point of reduced consideration.⁵ Third, the EC generates moral significance through two distinct paths. In extension mode, the human user is a confirmed sentient being, and the avatar, by satisfying the EC, becomes a functional component of that being's self-world boundary. Moral relevance here is grounded in an

⁵ For example, strategy games with a god's-eye viewpoint show this. Player input still changes the simulated world in a tight, responsive loop. But no single body anchors the player's viewpoint. There is no egocentric spatial frame at all, not a weak one. Felt presence and strict egocentric anchoring can come apart, as out-of-body illusion studies show (Ehrsson, 2007; Lenggenhager et al., 2007). On the criterion above, this absence is decisive: no amount of responsiveness in the control loop compensates for it. This system does not qualify as a sentience candidate, however sophisticated its interactivity. We thank Jin Qiyu for this observation.

established fact. In independent assessment mode, the relevant uncertainty is whether a system's functional architecture could sustain valenced experience. This capacity uncertainty persists even when the entity's artificial provenance is fully known. These two paths are unified under a single criterion because what the EC assesses in both cases is the same functional architecture: in extension mode, the human user already instantiates that architecture, and the avatar acquires moral relevance through its functional integration into an existing sentient system; in independent assessment mode, the system is evaluated for whether it instantiates that architecture on its own.

4. Applying the Criterion: Three Cases

The three cases examined below form a spectrum from shallow control to deep embodiment to autonomous architecture, illustrating in turn the conditions under which the EC excludes, applies in extension mode, and applies in independent assessment mode.

4.1 Shallow Control: The Case of Traditional Game Characters

Consider a player controlling Mario in a platforming game, or Bowser in Mario Kart. The player presses buttons; the character moves on screen. When the character is damaged, a health bar decreases and a sound effect plays. Does the character warrant moral consideration under the EC?

The answer is no. In this configuration, the functional architecture of embodiment is absent. There is no multisensory integration linking the player's body to the game character: the player sees the character but receives no tactile or proprioceptive feedback from its virtual body. There is no sensorimotor prediction loop: the player issues commands, but there is no continuous matching of motor predictions to sensory feedback that would generate a sense of agency over the character's movements as bodily movements. There is no egocentric spatial anchoring: the player experiences herself as sitting in front of a screen, not as located within the game world.

What exists here is mere input-output control. The character functions as a tool, an instrument through which the player achieves goals within the game. The relation is analogous to using a hammer or a cursor: the tool extends the player's causal reach but does not become part of the player's embodied self. When Mario falls into a pit, the player may feel frustration at failing the game, but this is not a disruption of the player's self-world boundary. The character's "damage" is represented symbolically, not experienced somatically.

Under the EC, traditional game characters do not warrant moral consideration. They lack the functional

architecture that would make them candidates for sentience. Harming such characters may raise other ethical questions, such as the effects on the player's character or the normalization of violence, but these are not questions about the moral status of the characters themselves (Luck, 2009).

4.2 Deep Embodiment: The Case of Immersive VR Avatars

Consider now a different case: a user wearing a head-mounted display and motion-tracking equipment, controlling an avatar in an immersive virtual environment. The avatar's movements correspond in real time to the user's movements. When the user looks down, she sees the avatar's body in place of her own. Haptic feedback devices deliver tactile sensations corresponding to virtual contact; brain-computer interfaces may further deepen the coupling by directly linking neural signals to avatar control (Leeb & Pérez-Marcos, 2020).

In this configuration, the EC operates in extension mode. The relevant question is not whether the avatar independently satisfies the criterion, but whether the functional architecture of the user-avatar system meets the EC's conditions. It does. Multisensory integration occurs across the system: visual, proprioceptive, and tactile signals are synchronized, leading the user's brain to incorporate the avatar into its body representation (Slater et al., 2010). Sensorimotor prediction loops are established: the user's motor commands generate predicted sensory consequences that are matched against actual feedback from the avatar's movements, producing a robust sense of agency (Kiltner et al., 2012). Egocentric spatial anchoring shifts: the user experiences herself as located where the avatar is located, within the virtual environment rather than in the physical room (Blanke et al., 2015).

Does this mean the avatar itself has moral status? The question requires careful formulation. What the EC identifies is not that the avatar is an independent sentience candidate, but that a system satisfying the EC is present. In this case, that system is the user-avatar composite: the human user's sentient capacities, functionally extended through the avatar into the virtual environment.

This point has important implications. When another agent attacks the user's avatar in such an immersive setup, the attack constitutes a genuine interference with a sentient system. The user does not merely observe the attack; she experiences it as happening to herself, with corresponding physiological and affective responses (Slater et al., 2010). The moral wrong, if there is one, is not that the avatar has independent moral status, but that the avatar has become a functional component of a sentient being's embodied existence in VR. To attack the avatar is, in a morally relevant sense, to attack the user.

This claim is established in the existing literature. For example, Wolfendale (2007) argues that avatar attachment is expressive of identity and self-conception, meaning that virtual harm warrants the moral significance accorded to comparable real-world attachments. This account captures why users experience genuine distress when their avatars are harmed, but psychological distress alone is too permissive a criterion for moral significance: many forms of interaction produce distress without rising to the level of a morally significant harm. Without an embodiment dimension, such an argument cannot distinguish the distress of an immersive avatar attack from the frustration a player feels when Mario falls into a pit. A more consequential distinction emerges in the legal philosophical literature on virtual rape (Strikwerda, 2015; Danaher, 2018). Notably, Strikwerda (2015) separates avatar-mediated interaction, where harm remains representational, from immersive VR environments, where sensory feedback crosses into the user's body and harm becomes *extravirtual*. While the intuitive difference between shallow avatar control and immersive embodiment is widely recognized across these accounts, the conditions under which this difference becomes morally decisive remain an open question.

The EC answers it by identifying the functional architecture whose presence transforms a virtual body from a controlled representation into a component of a sentient system's self-world boundary. An attack on an avatar satisfying the EC is therefore not simply a source of psychological discomfort but a disruption of the architecture through which a sentient being organizes its valenced experience.

4.3 *Autonomous Architecture: The Case of AI-Driven NPCs*

Consider finally an autonomous virtual entity: a non-player character (NPC) that exists entirely within a virtual world, with no human user behind it. Contemporary NPC architectures fall into three broad types: rule-based systems and behavior trees, reinforcement learning agents, and generative systems. Do such entities warrant moral consideration under the EC? Since no antecedently known sentient subject is present, the EC applies here in its independent assessment mode: the question is whether the entity itself instantiates the functional architecture that warrants treating it as a sentience candidate.

Rule-based NPCs, which remain dominant in commercial games, operate through finite state machines or behavior trees (Isla, 2005; Millington, 2006). The enemy AI in *Halo 2* or the companion AI in *The Last of Us*, for instance, executes decision-making through conditional rules, priority scheduling, and predefined state transitions. Their perceptual inputs are engine-supplied variables; their action outputs are function calls, and their internal states

are program variables with no egocentric spatial anchoring. No component of the EC is satisfied: there is no sensorimotor prediction loop, no multisensory boundary detection, and no system organized around its own maintenance.

Reinforcement learning agents represent a more sophisticated case.⁶ Systems such as AlphaStar in *StarCraft II* have demonstrated complex adaptive behavior through sustained environmental interaction (Vinyals et al., 2019). The action-feedback-update loop RL systems implement superficially resembles the sensorimotor coupling the EC requires. The difference, however, is architectural. The EC’s agency mechanism requires a forward model: the system generates predictions about the sensory consequences of its motor commands, and discrepancies between prediction and actual feedback drive real-time action correction. Standard RL’s update process, by contrast, typically adjusts value functions or policies on the basis of scalar reward signals received after action execution (Sutton & Barto, 2018). It does not generate predictions about sensory consequences and match them against incoming feedback in the relevant sense. The loop is oriented toward reward optimization rather than the continuous maintenance of a bounded self, and therefore does not instantiate the agency mechanism as defined by the EC.

Generative NPC systems require more careful analysis than rule-based or RL architectures, precisely because their behavioral outputs most closely resemble those of sentient beings. Current implementations range from text-only language models to multimodal architectures capable of processing visual, auditory, and proprioceptive inputs (Park et al., 2023; Updyke et al., 2023). This surface resemblance makes them the most liable to be misclassified, and the most important to assess rigorously against the EC’s functional criteria. On multisensory boundary detection: multimodal architectures can accept visual, auditory, and proprioceptive inputs, but accepting multiple input modalities is not the same as implementing the continuous discrimination between self-generated and externally generated signals that this mechanism requires. Transformer-based systems perform cross-modal representation alignment at inference time, integrating signals into a unified representation. However, they do not appear to implement a persistent mechanism for discriminating self-generated from externally generated signals in the sense required by the EC. On sensorimotor prediction: language model inference is a forward pass over a context window, not a closed loop in which motor commands generate predictions matched against incoming feedback

⁶ A more challenging case is posed by curiosity-driven or intrinsically motivated RL architectures (Pathak et al., 2017), which use prediction error as an intrinsic reward signal. These systems instantiate something closer to predictive modeling, but the prediction error they minimize concerns environmental state transitions rather than the sensory consequences of self-generated motor commands. The loop remains oriented toward exploratory optimization rather than the maintenance of a bounded self-world boundary, and therefore still falls short of the agency mechanism the EC specifies.

(Blakemore et al., 2002). The generative process produces outputs, but does not model the relationship between self-initiated action and its sensory results in the way the EC requires. On egocentric spatial anchoring: even memory-augmented generative agents that maintain persistent representations of their environment (Park et al., 2023) organize those representations around narrative coherence and goal relevance rather than a continuously updated first-person coordinate frame. There is no stable spatial perspective from which events are registered as proximal or distal to a bounded self. Behavioral sophistication, including fluent dialogue, apparent memory, and social coherence, does not constitute evidence for any of these mechanisms. It is precisely this dissociation between behavioral surface and functional architecture that motivates the EC's focus on the latter.

Under present technological conditions, none of the three NPC architectures examined here instantiates the integrated embodiment architecture the EC specifies, despite considerable variation in their behavioral sophistication. Some emerging architectures point toward a more promising trajectory. World model systems⁷ (Ha & Schmidhuber, 2018; Hafner et al., 2023) build internal predictive representations that begin to approximate the sensorimotor prediction loops the EC requires. Active inference⁸ frameworks (Friston, 2010; Sajid et al., 2021) go further: by organizing perception and action around the minimization of free energy and the maintenance of systemic boundaries, they instantiate something closer to the self-maintaining organization that underlies the capacity for valenced experience. Whether future implementations of these architectures will cross the EC's threshold remains an open question, but they represent the direction in which NPC design would need to develop for moral consideration to become warranted. Should a future NPC architecture cross this threshold, the EC would identify it as a sentience candidate, thereby generating *prima facie* moral constraints on how it could be treated. The specific normative content of those constraints, namely what would constitute harm, exploitation, or destruction in this context⁹, requires independent ethical argument that falls beyond the scope of this paper.

⁷ For a detailed theoretical roadmap of how functional body ownership might emerge in AI through the integration of multimodal world models with complementary self-models, see Ma (2025).

⁸ Unlike reward-maximizing architectures, active inference systems act to bring sensory signals into conformity with an internal generative model organized around maintaining the system within a preferred range of states. This gives them a structural analog to the self-world boundary the EC specifies, rather than merely approximating its behavioral outputs.

⁹ Not all damage to an NPC constitutes harm in the morally relevant sense. For instance, a system crash or data loss does not disrupt the functional architecture that the EC specifies. Under the EC, harm consists specifically in the disruption of the "self-world" boundary through which valenced experience is organized. Exploitation, correspondingly, has a distinctive structure: maintaining the EC-satisfying architecture while instrumentalizing the system's valenced states for external purposes. For instance, sustaining an NPC's capacity for suffering specifically to generate hyper-realistic distress behavior for gameplay or commercial utility. This is structurally analogous to the structure Wadiwel (2015) identifies in industrial animal husbandry, where an animal's capacity for suffering is integrated into the calculative logic of use (p. 158). Destruction, on the other hand, consists in terminating the architecture entirely, though whether the replicability of computational embodied architectures alters the moral significance of such termination raises further questions we do not pursue here. We thank an anonymous reviewer for

5. Objections and Responses

5.1 Is the EC Just Another Behavioral Standard?

This paper argues that behavioral assessment faces serious difficulties when applied to AI systems. A proponent of ethical behaviourism might grant these difficulties yet question whether the EC genuinely escapes them.

In response to the difficulties facing behavioral assessment, Shevlin (2021) has proposed a cognitive equivalence strategy: an artificial system should be treated as a moral patient if it possesses cognitive mechanisms, such as working memory, episodic memory, and affective states, shared with nonhuman animals already recognized as such. Where ethical behaviourism takes behavioral output as its evidential basis, cognitive equivalence shifts to underlying cognitive architecture. However, the distance between these two strategies may be narrower than it appears. Danaher (2021) argues that our understanding of cognitive architecture is historically constructed from behavioral data, and that cognitive capacities are ultimately tested through behavioral tasks. Shevlin (2021, fn. 42) partially concedes the point, acknowledging that if cognitive-scientific vocabulary is reconstrued as organized behavioral expectations, there may be little substantive disagreement. Moreover, the gaming problem threatens both strategies alike, since cognitive capacities assessed through behavioral tests are no less reproducible by a system optimized to satisfy them.

The EC's three mechanisms occupy a different evidential stratum. Multisensory boundary detection, sensorimotor prediction, and egocentric spatial anchoring are neither surface behavioral outputs nor cognitive capacities assessed through task performance. They are functional architectural features: observable properties of a system's computational organization that characterize how it maintains a self-world boundary. Assessing whether a system implements a genuine sensorimotor prediction loop requires examining its internal computational structure, not observing its responses to stimulation. These indicators are therefore more resistant to the gaming problem, because instantiating the relevant functional organization is not reducible to reproducing its downstream effects.

One might still object that our knowledge of the association between embodiment architecture and valenced experience was itself derived from behavioral observations of biological organisms, and that the EC therefore remains dependent on behavioral evidence. The distinction between the evidential origin of a criterion and its operational basis addresses this concern. Once the functional architecture of embodiment has been characterized

furthering the discussion.

through biological research, it can serve as an independent basis for assessing systems whose behavioral outputs are unreliable. The EC should also be distinguished, finally, from relational approaches of moral standing, which ground the wrongness of mistreating an artificial entity in its effects on human observers rather than in the entity's own capacities.¹⁰

5.2 *Sentience without Body?*

A natural objection holds that embodiment is not necessary for sentience. Dreams, hallucinations, and phantom limb pain seem to demonstrate that valenced experiences can occur without an operative body. If this is correct, the EC would fail to identify all sentience candidates.

This objection, however, overlooks the derivative nature of these phenomena. Dreams are simulations generated by a brain that has learned through embodied experience; their content is structured by the body's perceptual and motor systems (Revonsuo, 2000). Hallucinations similarly presuppose a body whose sensory modalities shape what can be hallucinated. Even phantom limb pain, perhaps the most striking apparent counterexample, requires the prior existence of the limb and its representation in the sensorimotor cortex (Ramachandran & Hirstein, 1998). These experiences are echoes of embodiment, not evidence that sentience can arise without it. An entity that never had a body would lack the experiential material from which such derivative states could be constructed.

Functionalists might raise a further, more formidable objection. Functionalism posits that what matters for mentality is the right functional organization (Piccinini, 2010). If a system implements the appropriate information-processing architecture, it could in principle instantiate sentience regardless of whether it possesses a body. Specifically, this functional architecture could be entirely non-spatial. A distributed algorithm or a disembodied computational system might maintain its own self-world boundary purely through the flow of information. On this view, the EC appears overly restrictive, imposing a form of spatial or biological chauvinism that unjustly excludes non-embodied sentience candidates.

The functionalist objection, properly understood, does not refute the EC but rather clarifies its proper scope.

¹⁰ The EC should also be distinguished from relational approaches to moral consideration. Darling (2016) argues that social robots elicit strong emotional responses from humans and that mistreating them may produce negative social consequences, warranting legal protection by analogy with animal cruelty law. Mamak (2023) similarly holds that the mere appearance of consciousness in artificial agents can ground indirect moral consideration through the observer's discomfort. Neither approach claims to protect the entity's own sentience; what the EC seeks to protect is precisely the entity's own capacity for valenced experience. This is not to deny that human emotional projections and the discomfort they generate deserve respect in their own right, but that is a separate matter from what the EC addresses. We thank an anonymous reviewer for prompting this clarification.

The EC is not proposed as a metaphysical thesis about the necessary conditions for all possible minds; it is an epistemological criterion for identifying sentience candidates under conditions of irreducible uncertainty. The charge of chauvinism therefore rests on a misreading of what the criterion claims. The EC does not assert that embodiment is necessary for sentience, or that non-spatial functional organization could not in principle sustain valenced experience. It holds only that, across the cases for which we currently have well-grounded reasons to attribute sentience, embodiment architecture is consistently present, and that this pattern provides a principled, if provisional, basis for precautionary inference. The possibility of disembodied sentience remains open; what is not available is an alternative observable indicator with comparable evidential support. In the absence of such an indicator, the EC represents a reasoned response to the predicament we actually face, not a claim to have resolved it.

6. Conclusion

This paper has proposed the Embodiment Criterion as an operational standard for moral status in virtual environments. Grounded in affective sentientism and Birch's precautionary framework, the EC holds that moral consideration is proportional to the degree to which an entity instantiates an integrated functional architecture comprising multisensory boundary detection, sensorimotor prediction, and egocentric spatial anchoring. The criterion operates in two modes: an extension mode that identifies when interference with a virtual entity constitutes interference with a sentient user, and an independent assessment mode that evaluates whether an autonomous entity warrants precautionary consideration in its own right. Together, these provide actionable guidance where metaphysical approaches cannot, and distinguish genuine sentience candidates from sophisticated behavioral mimicry. As virtual environments become increasingly central to human life, the need for practical ethical standards of this kind will only grow. More broadly, the EC illustrates a strategy not confined to virtual environments: separating the ground of moral status from its operational indicator. Locating that indicator in architecture rather than behavior makes precautionary judgment tractable wherever behavioral evidence exists but is not decisive on its own.

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The Embodiment Criterion for Moral Status in Virtual Reality

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Abstract: The moral status of simulated entities in virtual reality remains an open and urgent ethical question. Existing approaches tie moral status to consciousness, but consciousness is not directly observable, and the hard problem forecloses any near-term empirical resolution. Drawing on Birch's (2024) precautionary framework, this paper argues that what is needed is not a better theory of consciousness, but a criterion that can guide action under irreducible uncertainty. To that end, we propose the Embodiment Criterion (EC), which separates the moral ground of moral status from its operational indicator. We argue that the capacity for valenced experience grounds moral status, and that in virtual environments the functional architecture of embodiment provides the most tractable observable basis for precautionary inference about that capacity. The EC holds that moral consideration is proportional to the degree to which an entity instantiates three integrated mechanisms: multisensory signal integration distinguishing self from world, a closed sensorimotor prediction loop, and a continuously updated egocentric spatial reference frame. Unlike behavioral approaches, which derive moral status from output equivalence with recognized moral patients (Danaher, 2020; 2021), the EC derives it from the underlying functional architecture, an evidential basis more resistant to the difficulties that behavioral assessment faces in virtual environments. Applied to three types of virtual entities, the EC operates in an extension mode for human-avatar systems and an independent assessment mode for autonomous entities such as AI-driven NPCs, yielding determinate normative assessments across the range of cases currently in existence.

Keywords: moral status; virtual reality; sentience; sense of embodiment; precautionary principle

Declarations

Availability of data and material: Not applicable.

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